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Department of Mathematics

LECTURE NOTES

Advanced Calculus and Numerical Methods (21MAT21)

Module-3 ( Partial Differential Equations)

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### Partial Differential Equation:-

An equation which involves one dependent variable with respect to two or more independent variables is called partial differential equation.

eg:-  $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 3z$

### Notations:-

Let  $z = f(x, y)$  be a real valued continuous function defined by on an any other interval, then the first and second order partial derivatives of the function is notated as

1.  $\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x = p$

2.  $\frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y = q$

3.  $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 f}{\partial x^2} = z_{xx} = f_{xx} = r$

$$4. \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y} = z_{xy} = f_{xy} = s$$

$$5. \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 f}{\partial y^2} = z_{yy} = f_{yy} = t$$

Formation of PDE by elimination of arbitrary constant.

1. Form PDE by eliminating arbitrary constant from  $z = (x-a)^2 + (y-b)^2$ .

Sol:- Given,

$$z = (x-a)^2 + (y-b)^2 \longrightarrow (1)$$

diff. (1) partially w.r.t.  $x$

$$(1) \Rightarrow \frac{\partial z}{\partial x} = 2(x-a)(1) + 0$$

$$\frac{\partial z}{\partial x} = 2(x-a)$$

$$p = 2(x-a)$$

$$(x-a) = \frac{p}{2}$$

diff (1) p. w.r.t.  $y$

$$(1) \Rightarrow \frac{\partial z}{\partial y} = 0 + 2(y-b)(1)$$

$$\frac{\partial z}{\partial y} = 2(y-b)$$

$$q = 2(y-b)$$

$$(y-b) = \frac{q}{2}$$

$$(1) \Rightarrow z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 \Rightarrow \boxed{4z = p^2 + q^2}$$

2. Form PDE by eliminating arbitrary constant  $a, b$  from  
from  $(x-a)^2 + (y-b)^2 + z^2 = r^2$ .

Sol:

$$(x-a)^2 + (y-b)^2 + z^2 = r^2 \longrightarrow (1)$$

diff (1) p. w. r. to  $x$

$$\Rightarrow 2(x-a) + 0 + 2z \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow (x-a) + z \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow (x-a) = -z p \longrightarrow (2)$$

diff (1) p. w. r. to  $y$

$$\therefore (1) \Rightarrow 0 + 2(y-b) + 2z \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow (y-b) + z \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow (y-b) + z q = 0$$

$$(y-b) = -z q \longrightarrow (3)$$

$$\therefore (1) \Rightarrow (-pz)^2 + (-qz)^2 + z^2 = r^2$$

$$\Rightarrow p^2 z^2 + q^2 z^2 + z^2 = r^2$$

$$\Rightarrow (p^2 + q^2 + 1) z^2 = r^2$$

$$\Rightarrow (p^2 + q^2 + 1) z^2 = r^2$$

3. Form the PDE by eliminating arbitrary constants  $a, b$   
from  $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ .

Sol: Given,

$$2z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \longrightarrow (1)$$



Diff p. w. r. to  $x$

$$\frac{\partial z}{\partial x} = \frac{x}{a^2} + 0$$

$$\frac{\partial z}{\partial x} = \frac{x}{a^2}$$

$$p = \frac{x}{a^2}$$

$$a^2 = \frac{x}{p} \rightarrow (2)$$

Diff p. w. r. to  $y$

$$\Rightarrow \frac{\partial z}{\partial y} = 0 + \frac{y}{b^2}$$

$$\Rightarrow \frac{\partial z}{\partial y} = \frac{y}{b^2}$$

$$\Rightarrow q = \frac{y}{b^2}$$

$$\Rightarrow b^2 = \frac{y}{q}$$

$$(1) \Rightarrow z = \frac{x^2}{x/p} + \frac{y^2}{y/q}$$

$$\Rightarrow z = \frac{px^2}{x} + \frac{qy^2}{y}$$

$$\Rightarrow \boxed{z = px + qy}$$

4. Form PDE by eliminating arbitrary constants from

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Sol:-

$$\text{Let } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \longrightarrow (1)$$

diff p. w. r. to x

$$(1) \Rightarrow \frac{\partial x}{\partial x} + 0 + \frac{\partial z}{\partial x} \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{x}{a^2} + \frac{pz}{c^2} = 0 \longrightarrow (2)$$

diff p (2) w. r. to x

$$\Rightarrow \frac{1}{a^2} + \frac{1}{c^2} \left[ p \frac{\partial z}{\partial x} + z \frac{\partial p}{\partial x} \right] = 0$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{c^2} \left[ p \cdot p + z \cdot \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) \right] = 0$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{c^2} \left[ p^2 + z \frac{\partial^2 z}{\partial x^2} \right] = 0$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{c^2} [p^2 + \cancel{zx}] = 0 \longrightarrow (3)$$

$$\therefore (2) \Rightarrow \frac{x}{a^2} = -\frac{pz}{c^2}$$

$$\Rightarrow \frac{1}{a^2} = -\frac{pz}{c^2 x}$$

$$\therefore (3) \Rightarrow -\frac{pz}{c^2 x} + \frac{1}{c^2} [p^2 + \cancel{zx}] = 0$$

$$\Rightarrow -\frac{pz}{x} + (p^2 + \cancel{zx}) = 0$$

$$\Rightarrow \boxed{\frac{pz}{x} = p^2 + \cancel{zx}}$$

5. Form PDE by eliminating arbitrary function  $z = f(x^2 + y^2)$ .

Sol: Given,  $z = f(x^2 + y^2) \longrightarrow (1)$

diff (1) p w. r. t. x

$$(1) \Rightarrow \frac{\partial z}{\partial x} = f'(x^2 + y^2)(2x)$$

$$p = 2x f'(x^2 + y^2) \longrightarrow (2)$$

$$\text{||y, } \frac{\partial z}{\partial y} = f'(x^2 + y^2)(2y)$$

$$q = 2y f'(x^2 + y^2) \longrightarrow (3)$$

$$\Rightarrow \frac{(2)}{(3)} \Rightarrow \frac{p}{q} = \frac{\cancel{2x} f'(\cancel{x^2} + y^2)}{\cancel{2y} f'(\cancel{x^2} + y^2)}$$

$$\Rightarrow \frac{p}{q} = \frac{x}{y}$$

$$\Rightarrow py = qx$$

$$\Rightarrow \boxed{py - qx = 0}$$

6. Form PDE by eliminating arbitrary functions from  $ax + by + cz = f(x^2 + y^2 + z^2)$ .

Sol: Given,

$$ax + by + cz = f(x^2 + y^2 + z^2) \longrightarrow (1)$$

diff (1) p, w. r. t. x

$$(1) \Rightarrow a + 0 + c \frac{\partial z}{\partial x} = f'(x^2 + y^2 + z^2)(2x + 0 + 2z \frac{\partial z}{\partial x})$$

$$\Rightarrow a + cp = f'(x^2 + y^2 + z^2) 2(x + zp)$$

$$\Rightarrow a + pc = 2(x + zp) f'(x^2 + y^2 + z^2) \longrightarrow (2)$$



(1)  $\Rightarrow$  diff p. w. r. t. y

$$0 + b + c \frac{\partial z}{\partial y} = f'(x^2 + y^2 + z^2) (0 + 2y + 2z \frac{\partial z}{\partial y})$$

$$\Rightarrow b + cq = f'(x^2 + y^2 + z^2) 2(y + zq)$$

$$\Rightarrow b + qc = 2(y + zq) f'(x^2 + y^2 + z^2) \rightarrow (3)$$

$$\frac{(2)}{(3)} \Rightarrow \frac{a + pc}{b + qc} = \frac{2(x + zp) f'(x^2 + y^2 + z^2)}{2(y + zq) f'(x^2 + y^2 + z^2)}$$

$$\Rightarrow (a + pc)(y + zq) = (b + qc)(x + zp)$$

$$\Rightarrow (a + pc)(y + zq) - (b + qc)(x + zp) = 0$$

∴ Form PDE by eliminating arbitrary function from  $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$ .

Sol. Given,

$$x = f(x + ay) + q$$

$$z = y^2 + 2f\left(\frac{1}{x} + \log y\right) \rightarrow (1)$$

diff p. w. r. t. x

$$\Rightarrow \frac{\partial z}{\partial x} = 0 + 2f'\left(\frac{1}{x} + \log y\right) \left(-\frac{1}{x^2} + 0\right)$$

$$\Rightarrow p = -\frac{2}{x^2} f'\left(\frac{1}{x} + \log y\right) \rightarrow (2)$$

diff p. w. r. to y

$$\therefore (1) \Rightarrow \frac{\partial z}{\partial y} = 2y + 2f'\left(\frac{1}{x} + \log y\right) \left(0 + \frac{1}{y}\right)$$

$$q = 2y + \frac{2}{y} f'\left(\frac{1}{x} + \log y\right)$$

$$q - 2y = \frac{2}{y} f'\left(\frac{1}{x} + \log y\right) \rightarrow (3)$$

$$\frac{(2)}{(3)} \Rightarrow \frac{p}{q-ay} = \frac{-2/x^2 f'(\frac{1}{x} + \log y)}{2/y f'(\frac{1}{x} + \log y)}$$

$$\Rightarrow \frac{p}{q-ay} = -\frac{y}{x^2}$$

$$\Rightarrow px^2 = -y(q-ay)$$

$$\Rightarrow px^2 = -yq + ay^2$$

$$\Rightarrow px^2 + yq = ay^2.$$

8 Form PDE by eliminating arbitrary function

$$z = f(x+ay) + g(x-ay).$$

Sol: Given,

$$z = f(x+ay) + g(x-ay) \rightarrow (1)$$

diff (1) p. w. r. t x

$$(1) \Rightarrow \frac{\partial z}{\partial x} = f'(x+ay)(1) + g'(x-ay)(1)$$

$$\Rightarrow \frac{\partial z}{\partial x} = f'(x+ay) + g'(x-ay) \rightarrow (2)$$

diff. p (2) w. r. t x

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = f''(x+ay)(1) + g''(x-ay)(1) \rightarrow (3)$$

$$\text{||y} \quad \frac{\partial z}{\partial y} = f'(x+ay)(a) + g'(x-ay)(-a)$$

$$\Rightarrow \frac{\partial z}{\partial y} = af'(x+ay) - ag'(x-ay)$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = af''(x+ay)(a) - g''(x-ay)(a)$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = a^2 f''(x+ay) + a^2 g''(x-ay)$$



$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = a^2 [f''(x+ay) + g''(x-ay)]$$

$$\Rightarrow \boxed{\frac{\partial^2 z}{\partial y^2} = a^2 \frac{\partial^2 z}{\partial x^2}}$$

9. Form PDE by eliminating arbitrary functions

$$z = y f(x) + x \phi(y).$$

Sol: Given,

$$z = y f(x) + x \phi(y) \rightarrow (1)$$

diff. p. w. r. t. x

$$\Rightarrow \frac{\partial z}{\partial x} = y f'(x) + (1) \phi(y)$$

$$\Rightarrow \frac{\partial z}{\partial x} = y f'(x) + \phi(y)$$

$$\Rightarrow y f'(x) = p - \phi(y)$$

$$\Rightarrow f'(x) = \frac{p - \phi(y)}{y} \rightarrow (2)$$

$$\text{Illy } \frac{\partial z}{\partial y} = 1 f(x) + x \phi(y)$$

$$\Rightarrow \frac{\partial z}{\partial y} = f(x) + x \phi(y) \rightarrow (3)$$

$$\Rightarrow x \phi(y) = q - f(x)$$

$$\Rightarrow \phi(y) = \frac{q - f(x)}{x} \rightarrow (4)$$

diff. p. (3) w. r. t. x

$$\therefore (3) \Rightarrow \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) = f'(x) + 1 \phi'(y)$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = f'(x) + \phi'(y)$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = \frac{p - \phi(y)}{y} + \frac{q - f(x)}{x}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = \frac{x(p - \phi(y)) + y(q - f(x))}{xy}$$

$$\Rightarrow xy \cdot \frac{\partial^2 z}{\partial x \partial y} = px - x\phi(y) + qy - yf(x)$$

$$\Rightarrow xy \cdot \frac{\partial^2 z}{\partial x \partial y} = px + qy - [yf(x) + x\phi(y)]$$

$$\Rightarrow xy \cdot \frac{\partial^2 z}{\partial x \partial y} = px + qy - z \quad (\because \text{from eq.-(1)})$$

$$\Rightarrow xys = px + qy - z$$

$$\Rightarrow \boxed{px + qy - xys - z = 0}$$

Formation of PDE for composite functions:-

1. Let  $\phi(u, v) = 0$  be a composite function for which  $u = f(x, y, z)$ ,  $v = f(x, y, z) \rightarrow (1)$
2. By differentiating  $\phi$  partially with respect to 'x' we get,  $\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial x} = 0$   

$$\Rightarrow \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} = - \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial x} \rightarrow (2)$$
3. Similarly, when differentiating with respect 'y'.

We get,  $\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} = 0$

$$\Rightarrow \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} = - \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} \rightarrow (3)$$

$$\frac{(9)}{(3)} \Rightarrow \frac{\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x}}{\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y}} = \frac{- \frac{\partial \phi}{\partial v} \cdot \frac{\partial u}{\partial x}}{- \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y}}$$

$$\Rightarrow \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = \frac{\frac{\partial u}{\partial x}}{\frac{\partial v}{\partial y}}$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial v}{\partial x} \cdot \frac{\partial u}{\partial y} = 0}$$

1. Form PDE by eliminating arbitrary functions from  $f(x+y+z, x^2+y^2+z^2) = 0$

Sol:-  $f(x+y+z, x^2+y^2+z^2) = 0$

$$\Rightarrow f(u, v) = 0$$

$$u = x+y+z \quad v = x^2+y^2+z^2$$

$$\frac{\partial u}{\partial x} = 1+0+\frac{\partial z}{\partial x} \Rightarrow \frac{\partial u}{\partial x} = 1+p$$

$$\frac{\partial u}{\partial y} = 0+1+\frac{\partial z}{\partial y} \Rightarrow \frac{\partial u}{\partial y} = 1+q$$

$$\frac{\partial v}{\partial x} = 2x+0+2z\frac{\partial z}{\partial x} \Rightarrow \frac{\partial v}{\partial x} = 2(x+pz)$$



$$\frac{\partial v}{\partial y} = 0 + \cancel{xy} + \cancel{yz} \frac{\partial z}{\partial y} \Rightarrow \frac{\partial v}{\partial y} = \cancel{y} (y + qz)$$

WKT,

The required PDE eliminated by functions

$$\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow (1+p) \cancel{y} (y + qz) - \cancel{y} (x + pz) (1+q) = 0$$

$$\Rightarrow (1+p) (y + qz) - (1+q) (x + pz) = 0$$

$$\Rightarrow y + qz + py + pqz - x - pz - qx - pq = 0$$

$$\Rightarrow \boxed{(y-z)p + (z-x)q = x-y}$$

2. Form PDE by eliminating arbitrary function from  $z = f(x^2 + y^2, z - xy) = 0$ .

Sol:-

$$z = f(x^2 + y^2, z - xy) = 0 \rightarrow (1)$$

$$\Rightarrow f(u, v) = 0$$

$$u = x^2 + y^2, \quad v = z - xy$$

$$\frac{\partial u}{\partial x} = 2x \quad \frac{\partial v}{\partial x} = \frac{\partial z}{\partial x} - y = (p - y)$$

$$\frac{\partial u}{\partial y} = 2y \quad \frac{\partial v}{\partial y} = \frac{\partial z}{\partial y} - x = (q - x)$$

WKT,

The required PDE eliminated by function.

$$\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}$$

$$\Rightarrow (2x)(q-x) - (2y)(p-y) =$$

$$\Rightarrow x(q-x) - y(p-y) = 0$$

$$\Rightarrow qx - x^2 - py + y^2 = 0$$

$$\Rightarrow \boxed{qx - py = x^2 - y^2}$$

3. Form PDE by eliminating arbitrary function from  
 $z = f\left(\frac{xy}{z}, z\right) = 0$

Sol:-  $z = f\left(\frac{xy}{z}, z\right) = 0$

$$\Rightarrow f(u, v) = 0$$

$$u = \frac{xy}{z}, v = z$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{z(y) - xy \cdot \frac{\partial z}{\partial x}}{z^2} \quad \Rightarrow \frac{\partial v}{\partial x} = \frac{\partial z}{\partial x} = p$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{zy - xyp}{z^2}$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{z(x) - xy \cdot \frac{\partial z}{\partial y}}{z^2} \quad \Rightarrow \frac{\partial v}{\partial y} = \frac{\partial z}{\partial y} = q$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{zx - xypq}{z^2}$$

The required solution is

$$\Rightarrow \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow \left(\frac{zy - xyp}{z^2}\right) \left(\frac{\partial z}{\partial y}\right) - \left(\frac{zx - xypq}{z^2}\right) \left(\frac{\partial z}{\partial x}\right) = 0$$

$$\Rightarrow \left(\frac{zy - xyp}{z^2}\right) (q) - \left(\frac{zx - xypq}{z^2}\right) (p) = 0$$

$$\Rightarrow \frac{qzy - xypq}{z^2} = \frac{pzx - xypq}{z^2}$$

$$\Rightarrow qzy - pzx - xypq + xypq = 0$$

$$\Rightarrow (qy - px)z = 0$$

$$\Rightarrow (qy - px) = 0$$

Solution of PDE by direct Integration:-

1. Solve  $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$

Sol:  $\Rightarrow \frac{\partial}{\partial x} \left[ \frac{\partial^2 z}{\partial x \partial y} \right] = \cos(2x + 3y)$

$$\Rightarrow \int \frac{\partial^2 z}{\partial x \partial y} = \int \cos(2x + 3y) dx$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = \frac{\sin(2x + 3y)}{2} + f(y)$$

$$\Rightarrow \frac{\partial}{\partial y} \left[ \frac{\partial z}{\partial x} \right] = \frac{1}{2} \sin(2x + 3y) + f(y)$$

$$\Rightarrow \int \frac{\partial z}{\partial x} = \frac{1}{2} \int \sin(2x + 3y) dy + \int f(y) dy$$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{\cos(2x + 3y)}{6} + F(y) + g(x)$$

$$\Rightarrow \int dz = -\frac{1}{6} \int \cos(2x + 3y) dx + F(y) \int 1 dx + \int g(x) dx$$

$$\Rightarrow z = -\frac{1}{6} \frac{\sin(2x + 3y)}{2} + x F(y) + G(x) + h(y)$$

$$\Rightarrow z = -\frac{1}{12} \sin(2x + 3y) + x F(y) + G(x) + h(y)$$

where  $F(y) = \int f(y) dy$

$$G(x) = \int g(x) dx$$



Q. Solve  $\frac{\partial^2 z}{\partial x^2} = xy$ , subjects to condition that

$\frac{\partial z}{\partial x} = \log(1+y)$  at  $x=1$ , and  $z=0$  at  $x=0$

Sol: Given,  $\frac{\partial^2 z}{\partial x^2} = xy$

$$\Rightarrow \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) = xy \Rightarrow \partial \left( \frac{\partial z}{\partial x} \right) = xy dx$$

$$\Rightarrow \int \partial \left( \frac{\partial z}{\partial x} \right) = \int xy dx$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{x^2 y}{2} + f(y) \rightarrow (1)$$

$$\Rightarrow \partial z = \frac{x^2 y}{2} \partial x + f(y) \partial x$$

$$\Rightarrow \int \partial z = \frac{y}{2} \int x^2 dx + f(y) \int \partial x$$

$$\Rightarrow z = \frac{y}{2} \cdot \frac{x^3}{3} + x f(y) + g(y)$$

$$\Rightarrow z = \frac{x^3 y}{6} + x f(y) + g(y) \rightarrow (2)$$

Given,  $\frac{\partial z}{\partial x} = \log(1+y)$  at  $x=1$

$$(1) \Rightarrow \log(1+y) = \frac{y}{2} + f(y)$$

$$\Rightarrow f(y) = \log(1+y) - \frac{y}{2}$$

$$\therefore (2) \Rightarrow z = \frac{x^3 y}{6} + x \left[ \log(1+y) - \frac{y}{2} \right] + g(y) \rightarrow (3)$$

and given  $z=0$  at  $x=0$

$$0 = 0 + g(y)$$

$$\Rightarrow g(y) = 0$$

$$\therefore z = \frac{x^3 y}{6} + x \log(1+y) - \frac{xy}{9}$$

3. Solve  $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y}$  subject to condition  $\frac{\partial z}{\partial x} = \log x$ , when  $y=1$  and  $z=0$ , when  $x=1$ .

Sol:- Given  $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y}$

$$\Rightarrow \frac{\partial}{\partial y} \left[ \frac{\partial z}{\partial x} \right] = \frac{x}{y}$$

$$\Rightarrow \partial \left[ \frac{\partial z}{\partial x} \right] = \frac{x}{y} \partial y$$

$$\Rightarrow \int \partial \left[ \frac{\partial z}{\partial x} \right] = \int \frac{x}{y} \partial y$$

$$\frac{\partial z}{\partial x} = x \log y + f(x) \rightarrow (1)$$

$$\Rightarrow \int dz = \int x \log y \, dx + \int f(x) \, dx$$

$$\Rightarrow z = \frac{x^2}{2} \log y + F(x) + g(y) \rightarrow (2)$$

where  $F(x) = \int f(x) \, dx$

given  $\frac{\partial z}{\partial x} = \log x$  at  $y=1$

$$(1) \Rightarrow \log x = x \log 1 + f(x)$$

$$\log x = f(x)$$

$$\therefore F(x) = \int f(x) \, dx$$

$$= \int \log x \, dx$$

$$\therefore F(x) = x(\log x - 1)$$

$$\therefore (2) \Rightarrow z = \frac{x^2}{2} \log y + x(\log x - 1) + g(y) \rightarrow (3)$$

and given  $z=0$  at  $x=1$ .

$$\therefore (3) \Rightarrow 0 = \frac{1}{2} \log y + 1(0-1) + g(y)$$

$$\Rightarrow g(y) = 1 - \frac{1}{2} \log y$$

$$\Rightarrow g(y) = 1 - \log \sqrt{y}$$

$$\therefore z = \frac{x^2}{2} \log y + x \log x - x + 1 - \log \sqrt{y}$$

4. Solve  $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$  subject to the condition that  $\frac{\partial z}{\partial y} = -\frac{1}{2} \sin y$  when  $x=0$  &  $z=0$  when  $y$  is odd multiple of  $\pi/2$ .

Sol: Given,  $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$

$$\Rightarrow \frac{\partial}{\partial x} \left[ \frac{\partial z}{\partial y} \right] = \sin x \sin y$$

$$\Rightarrow \partial \left[ \frac{\partial z}{\partial y} \right] = \sin x \sin y \, dx$$

$$\Rightarrow \int \partial \left[ \frac{\partial z}{\partial y} \right] = \sin y \int \sin x \, dx$$

$$\Rightarrow \frac{\partial z}{\partial y} = \sin y (-\cos x) + f(y) \rightarrow (1)$$

$$\Rightarrow \frac{\partial z}{\partial y} = \sin y (-\cos x) + f(y)$$

$$\Rightarrow \partial z = -\cos x \sin y \, dy + f(x) \, dy$$

$$\Rightarrow \int dz = -\cos x \int \sin y \, dy + \int f(x) \, dy$$



$$\Rightarrow z = \cos x \cos y + F(y) + g(x) \rightarrow (2)$$

$$\text{where } F(y) = \int f(y) dy$$

$$\text{Given, } \frac{\partial z}{\partial y} = -\sin y \text{ at } x=0$$

$$\therefore (1) \Rightarrow -\sin y = -\cos 0 \sin y + f(y)$$

$$\Rightarrow -\sin y = -\sin y + f(y)$$

$$\Rightarrow f(y) = -\sin y + \sin y$$

$$\Rightarrow f(y) = -\sin y$$

$$\therefore F(y) = \int f(y) dy$$

$$\therefore F(y) = -\int \sin y dy$$

$$F(y) = \cos y$$

$$(2) \Rightarrow z = \cos x \cos y + \cos y + g(x) \rightarrow (3)$$

$$\text{and given } z=0, \text{ where } y = (2n+1)\frac{\pi}{2}$$

$$\therefore (3) \Rightarrow 0 = \cos x \cos (2n+1)\frac{\pi}{2} + \cos (2n+1)\frac{\pi}{2} + g(x)$$

$$\Rightarrow 0 + 0 + 0 + g(x)$$

$$\Rightarrow g(x) = 0$$

$\therefore$  The solution is

$$z = \cos x \cos y + \cos y$$

### Solution of Homogeneous PDE :-

1. Solve  $\frac{\partial^2 z}{\partial y^2} = z$ , given that  $y=0$ ,  $z=e^x$ ,  $\frac{\partial z}{\partial y} = e^{-x}$

Sol:- Given,  $\frac{\partial^2 z}{\partial y^2} = z$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} - z = 0$$

$$\Rightarrow (D^2 - 1)z = 0, \text{ where } D = \frac{\partial}{\partial y}$$

$$\therefore \text{AE is } m^2 - 1 = 0$$

$$\Rightarrow m^2 = 1$$

$$\Rightarrow m = \pm 1$$

The solution is

$$z = f(x)e^{-y} + g(x)e^y \rightarrow (1)$$

diff p (1) w.r.t  $y$

$$\therefore (1) \Rightarrow \frac{\partial z}{\partial y} = -f(x)e^{-y} + g(x)e^y \rightarrow (2)$$

given  $z = e^x$ , where  $y=0$

$$\therefore (1) \Rightarrow e^x = f(x) + g(x) \rightarrow (3)$$

$$\text{and } \frac{\partial z}{\partial y} = e^{-x}, \text{ where } y=0$$

$$\therefore (2) \Rightarrow e^{-x} = -f(x) + g(x) \rightarrow (4)$$

$$\therefore (3) + (4) \Rightarrow e^x + e^{-x} = 2g(x)$$

$$\Rightarrow g(x) = \frac{e^x + e^{-x}}{2}$$

$$\Rightarrow g(x) = \cosh x$$

$$(3) - (4) \Rightarrow e^x - e^{-x} = 2f(x)$$

$$\Rightarrow f(x) = \frac{e^x - e^{-x}}{2}$$

$$\Rightarrow f(x) = \sinh x$$

$$\therefore (1) \Rightarrow \boxed{z = e^{-y} \sinh x + e^y \cosh x}$$

Q. Solve  $\frac{\partial^2 z}{\partial x^2} + z = 0$ , given that when  $x = 0$ ,  $z = 0$  and  $\frac{\partial z}{\partial x} = 1$ .

Sol: Given,  $\frac{\partial^2 z}{\partial x^2} + z = 0$

$$\Rightarrow (D^2 + 1)z = 0$$

$$\text{where } D = \frac{\partial}{\partial x}$$

$\therefore$  The A.E is given by  $m^2 + 1 = 0$

$$m^2 = -1$$

$$\boxed{m = \pm i}$$

$$a = 0, b = 1$$

The solution is

$$z = e^{0 \cdot x} [f(y) \cos x + g(y) \sin x]$$

$$z = f(y) \cos x + g(y) \sin x \longrightarrow (1)$$

diff. p. w. r. t  $x$

$$\frac{\partial z}{\partial x} = -f(y) \sin x + g(y) \cos x \longrightarrow (2)$$

$$\therefore (1) \Rightarrow e^y = f(y) \cos 0 + g(y) \sin 0$$

$$\Rightarrow e^y = f(y)$$

$$\text{and given } \frac{\partial z}{\partial x} = 1 \text{ at } x = 0$$

$$\therefore (2) \Rightarrow 1 = -f(y) \sin 0 + g(y) \cos 0$$

$$1 = g(y)$$

$$\therefore g(y) = 1$$

$\therefore$  The solution is  $z = e^y \cos x + \sin x$



3. Solve  $\frac{\partial^2 z}{\partial x^2} = a^2 z$ , given that  $z=0$ ,  $\frac{\partial z}{\partial x} = a \sin y$  at  $x=0$ .

Sol: Given,  $\frac{\partial^2 z}{\partial x^2} - a^2 z = 0$

$$\Rightarrow (D^2 - a^2)z = 0$$

$$\text{where } D = \frac{\partial}{\partial x}$$

$$\therefore \text{The A.E is } (m^2 - a^2) = 0$$

$$\Rightarrow m^2 = a^2$$

$$\Rightarrow \boxed{m = \pm a} \quad m = -a, +a$$

The solution is

$$z = f(y)e^{-ax} + g(y)e^{ax} \longrightarrow (1)$$

diff (1) p. w. r. t  $x$

$$\frac{\partial z}{\partial x} = f(y)e^{-ax}(-a) + g(y)e^{ax}(a)$$

$$\frac{\partial z}{\partial x} = -af(y)e^{-ax} + ag(y)e^{ax} \longrightarrow (2)$$

$$\text{Given } \frac{\partial^2 z}{\partial x^2} = a \sin y \text{ at } x=0, z=0$$

$$\Rightarrow a \sin y = -af(y)e^{-a(0)} + ag(y)e^{a(0)}$$

$$\Rightarrow a \sin y = -af(y) + ag(y)$$

$$\Rightarrow a \sin y = a[-f(y) + g(y)]$$

$$\Rightarrow \sin y = -f(y) + g(y) \longrightarrow (4)$$

when  $z=0$ ,  $x=0$

$$(1) \Rightarrow 0 = f(y) + g(y) \longrightarrow (3)$$

$$(3) + (4) \Rightarrow \sin y = 2g(y)$$

$$\Rightarrow g(y) = \frac{\sin y}{2}$$

$$\Rightarrow f(y) = -g(y)$$

$$\Rightarrow f(y) = -\frac{\sin y}{2}$$

$$(3) - (4)$$

$$f(y) + g(y) - [g(y) - f(y)] = 0 - \sin y$$

$$f(y) + g(y) - g(y) + f(y) = -\sin y$$

$$2f(y) = -\sin y$$

$$f(y) = -\frac{\sin y}{2}$$

$$z = -\frac{\sin y}{2} e^{-ax} + \frac{\sin y}{2} e^{ax}$$

$$z = \sin y \left[ \frac{-e^{-ax}}{2} + \frac{e^{ax}}{2} \right]$$

$$z = \sin y \left[ \frac{-e^{-ax} + e^{ax}}{2} \right]$$

$$z = \sin y \left[ \frac{e^{ax} - e^{-ax}}{2} \right]$$

$$z = \sin y \sinh ax$$

4.  $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial z}{\partial x} - 4z = 0$  subject to conditions  $z=1$  and  $\frac{\partial z}{\partial x} = y$  when  $x=0$ .

Sol:- Given,  $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial z}{\partial x} - 4z = 0$

$$(D^2 + 3D - 4)z = 0$$

$$\text{where } D = \frac{\partial}{\partial x}$$

$$\Rightarrow m^2 + 3m - 4 = 0$$

$$m = -4, 1$$

$\therefore$  The solution is

$$z = e^{-4x} f(y) + e^x g(y) \longrightarrow (1)$$

diff. p. w. r. t  $x$

$$(1) \Rightarrow \frac{\partial z}{\partial x} = -4e^{-4x} f(y) + e^x g(y) \longrightarrow (2)$$

given  $z=1$ , when  $x=0$

$$(2) \Rightarrow 1 = e^0 f(y) + e^0 g(y)$$

$$\Rightarrow 1 = f(y) + g(y) \longrightarrow (3)$$

and given  $\frac{\partial z}{\partial x} = y$ , when  $x=0$

$$(2) \Rightarrow y = -4e^0 f(y) + e^0 g(y)$$

$$\Rightarrow y = -4f(y) + g(y) \longrightarrow (4)$$

$$(3) - (4)$$

$$1 - y = f(y) + g(y) + 4f(y) - g(y)$$

$$1 - y = 5f(y)$$

$$f(y) = \frac{1-y}{5}$$

$$\Rightarrow g(y) = 1 - f(y)$$

$$\Rightarrow g(y) = 1 - \left( \frac{1-y}{5} \right)$$

$$\Rightarrow g(y) = \frac{5-1+y}{5}$$

$$\Rightarrow g(y) = \frac{4+y}{5}$$

$$z = e^{-4x} \left( \frac{1-y}{5} \right) + e^x \left( \frac{4+y}{5} \right)$$



### Solution of PDE by Lagrange's Method:-

1. Let  $P = \frac{\partial z}{\partial x}$ ,  $Q = \frac{\partial z}{\partial y}$  then the equation  $Pp + Qq = R$

is called Lagrange's PDE, where  $P, Q, R$  are the functions in the variables of  $x, y, z$ .

2. Identify  $P, Q, R$  and write auxiliary equation

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

3. Solve the auxiliary equation and get the solutions

$$u(x, y, z) = C_1, \text{ \& } v(x, y, z) = C_2.$$

4. Write the solution of the given PDE in the form of  $\phi(u, v) = 0$ .

1. Solve  $(y-z)p + (z-x)q = x-y$ .

Sol:- Given,  $(y-z)p + (z-x)q = x-y$

$$\Rightarrow Pp + Qq = R$$

$$P = (y-z), Q = (z-x), R = x-y$$

$$\therefore \text{The A.E is } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y} = k \text{ (say)}$$

$$dx = (y-z)k$$

$$dy = (z-x)k$$

$$dz = (x-y)k$$

$$\text{Case-1: } dx + dy + dz = k(y-z) + k(z-x) + k(x-y)$$

$$\Rightarrow dx + dy + dz = k[y-z + z-x + x-y]$$

$$\Rightarrow dx + dy + dz = 0$$

$$\Rightarrow \int dx + \int dy + \int dz = 0$$

$$x + y + z = C_1$$

$$\therefore u(x, y, z) = x + y + z.$$

Case-2:- Let  $x dx + y dy + z dz = kx(y-z) + ky(z-x) + kz(x-y)$

$$\Rightarrow x dx + y dy + z dz = k [x\cancel{y} - x\cancel{z} + y\cancel{z} - x\cancel{y} + \cancel{z}x - \cancel{z}y]$$

$$\Rightarrow x dx + y dy + z dz = 0$$

$$\Rightarrow \int x dx + \int y dy + \int z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = 0C$$

$$\Rightarrow x^2 + y^2 + z^2 = 2C = C_2$$

$$\therefore u(x, y, z) = x^2 + y^2 + z^2$$

$\therefore$  The solution  $\phi(u, v) = 0$

$$\Rightarrow \phi(x + y + z, x^2 + y^2 + z^2) = 0$$

Q. Solve  $x(y-z)p + y(z-x)q = z(x-y)$

Sol. Given,  $\Rightarrow Pp + Qq = R$

$$\text{A.E is } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} = k$$

$$\Rightarrow dx = kx(y-z)$$

$$dy = ky(z-x)$$

$$dz = kz(x-y)$$

Case-1:-

$$\text{let } \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = k [y-z + z-x + x-y]$$

$$\Rightarrow \int \frac{1}{x} dx + \int \frac{1}{y} dy + \int \frac{1}{z} dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log C_1$$

$$\Rightarrow \log(xyz) = \log C_1$$

$$\Rightarrow xyz = C_1$$

$$\therefore u = xyz$$

Case 2:-

$$\text{Let } dx + dy + dz = kx(y-z) + ky(z-x) + kz(x-y)$$

$$\Rightarrow dx + dy + dz = k[x\cancel{y} - x\cancel{z} + y\cancel{z} - x\cancel{y} + z\cancel{x} - z\cancel{y}]$$

$$\Rightarrow dx + dy + dz = 0$$

$$\Rightarrow \int dx + \int dy + \int dz = 0$$

$$\Rightarrow x + y + z = C_2$$

$$\therefore v = x + y + z$$

$\therefore$  The solution is  $\phi(u, v) = 0$

$$\phi(xyz, x+y+z) = 0$$

3. Solve  $(y^2 - z^2)p + (z^2 - x^2)q = (x^2 - y^2)$

Sol:-  $\Rightarrow Pp + Qq = R$

$$\text{AE is } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{(y^2 - z^2)} = \frac{dy}{(z^2 - x^2)} = \frac{dz}{(x^2 - y^2)} = k$$

$$\Rightarrow dx = k(y^2 - z^2)$$

$$dy = k(z^2 - x^2)$$

$$dz = k(x^2 - y^2)$$



Case-1:

$$\text{Let } dx + dy + dz = k [y^2 - z^2 + z^2 - x^2 + x^2 - y^2]$$

$$\Rightarrow dx + dy + dz = 0$$

$$\Rightarrow \int dx + \int dy + \int dz = 0$$

$$\Rightarrow x + y + z = C_1$$

$$\therefore u = x + y + z$$

Case-2:-

$$\Rightarrow x^2 dx + y^2 dy + z^2 dz = k [x^2 y^2 - x^2 z^2 + y^2 z^2 - y^2 x^2 + z^2 x^2 - z^2 y^2]$$

$$\Rightarrow x^2 dx + y^2 dy + z^2 dz = 0$$

$$\Rightarrow \int x^2 dx + \int y^2 dy + \int z^2 dz = 0$$

$$\Rightarrow \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} = C$$

$$\Rightarrow x^3 + y^3 + z^3 = 3C = C_2$$

$$\therefore v = x^3 + y^3 + z^3$$

$\therefore$  The solution is  $\phi(u, v) = 0$

$$\Rightarrow \phi(x + y + z, x^3 + y^3 + z^3) = 0$$

4. Solve  $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$

Sol:  $\Rightarrow Pp + Qq = R$

A.E is  $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)} = k$$

$$\Rightarrow dx = kx(y^2 - z^2)$$

$$dy = ky(z^2 - x^2)$$

$$dz = kz(x^2 - y^2)$$

Case-1:-

$$\text{Let } \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = k [y^2 - z^2 + z^2 - x^2 + x^2 - y^2]$$

$$\Rightarrow \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$$

$$\Rightarrow \int \frac{1}{x} dx + \int \frac{1}{y} dy + \int \frac{1}{z} dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log C_1$$

$$\Rightarrow \log (xyz) = \log C_1$$

$$\Rightarrow xyz = C_1$$

$$\therefore u = xyz$$

Case-2:-

$$x dx + y dy + z dz = k [x^2/y^2 - x^2/z^2 + y^2/z^2 - y^2/x^2 + z^2/x^2 - z^2/y^2]$$

$$\int x dx + \int y dy + \int z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C$$

$$\Rightarrow x^2 + y^2 + z^2 = 2C = C_2$$

$$\therefore v = x^2 + y^2 + z^2$$

$\therefore$  The solution is  $\phi(u, v) = 0$

$$\phi(xyz, x^2 + y^2 + z^2) = 0$$

5.  $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)r$

Sol:-  $Pp + Qq = R$

A.E is  $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)} = k$$

$$dx = kx (y^2 - z^2)$$

$$dy = ky (z^2 - x^2)$$

$$dz = kz (x^2 - y^2)$$

$$\text{case-1:- } xdx + ydy + zdz = k \left[ \cancel{x^2/y^2} - \cancel{x^2/z^2} + \cancel{y^2/z^2} - \cancel{y^2/x^2} + \cancel{z^2/x^2} - \cancel{z^2/y^2} \right]$$

$$\Rightarrow xdx + ydy + zdz = 0$$

$$\Rightarrow \int xdx + \int ydy + \int zdz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C$$

$$\Rightarrow x^2 + y^2 + z^2 = 2C = C_1 \quad \therefore u = x^2 + y^2 + z^2$$

case-2:

$$\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = k \left[ \cancel{y^2/z^2} + \cancel{z^2/x^2} - \cancel{x^2/z^2} + \cancel{x^2/y^2} - \cancel{y^2/x^2} \right]$$

$$\Rightarrow \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = 0$$

$$\Rightarrow \int \frac{1}{x}dx + \int \frac{1}{y}dy + \int \frac{1}{z}dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log C_2$$

$$\Rightarrow \log(xyz) = \log C_2$$

$$\Rightarrow xyz = C_2 \quad \therefore v = xyz$$

The solution is  $\phi(u, v) = 0$

$$\Rightarrow \phi(x^2 + y^2 + z^2, xyz) = 0$$

6. Solve  $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Sol:-  $\Rightarrow Pp + Qq = R$

A.E is  $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = k$$



$$dx = x^2 k(y-z)$$

$$dy = y^2 k(z-x)$$

$$dz = z^2 k(x-y)$$

case-1:-

$$\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = k \left[ \frac{x^2}{y} - \frac{x}{z} + \frac{y}{z} - \frac{x}{y} + \frac{x}{z} - \frac{y}{z} \right]$$

$$\Rightarrow \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$$

$$\Rightarrow \int \frac{1}{x} dx + \int \frac{1}{y} dy + \int \frac{1}{z} dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log C_1$$

$$\Rightarrow \log (xyz) = \log C_1$$

$$\Rightarrow xyz = C_1 \quad \therefore u = xyz$$

case-2:-

$$\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz = k \left[ \frac{y^2}{z} + \frac{z}{x} - \frac{x}{y} + \frac{x}{z} - \frac{y}{z} \right]$$

$$\Rightarrow \int \frac{1}{x^2} dx + \int \frac{1}{y^2} dy + \int \frac{1}{z^2} dz = 0$$

$$\Rightarrow -\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = C$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -C \quad v = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

The solution is

$$\Rightarrow \phi(u, v) = 0$$

$$\phi \left( \frac{1}{x} + \frac{1}{y} + \frac{1}{z} ; xyz \right) = 0$$

7. Solve  $yzp + zxcq = xy$

Sol:  $\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$$

$$P = yz \quad Q = zx \quad R = xy$$

$$\text{Let } \frac{dx}{yz} = \frac{dy}{zx}$$

$$\int x dx = \int y dy$$

$$\frac{x^2}{2} = \frac{y^2}{2} + C_1$$

$$\frac{x^2}{2} - \frac{y^2}{2} = C$$

$$x^2 - y^2 = 2C = C_1$$

$$\boxed{u = x^2 - y^2}$$

$$\text{Let } \frac{dy}{zx} = \frac{dz}{xy}$$

$$\int y dy = \int z dz$$

$$\frac{y^2}{2} = \frac{z^2}{2} + C$$

$$\frac{y^2}{2} - \frac{z^2}{2} = C$$

$$y^2 - z^2 = 2C = C_2$$

$$\therefore \boxed{v = y^2 - z^2}$$

$\therefore$  The solution is  $\phi(u, v) = 0$

$$\Rightarrow \phi(x^2 - y^2, y^2 - z^2) = 0$$

8. Solve  $(y+z)p + (z+x)q = x+y$ .

Sol: Let  $(y+z)p + (z+x)q = x+y$

$$P = y+z \quad Q = z+x \quad R = x+y$$

$$\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$$

WKT,

$$\Rightarrow \frac{dx - dy}{y+z - z - x} = \frac{dy - dz}{z+x - x - y} = \frac{dz - dx}{x+y - y - z}$$

$$\Rightarrow \frac{dx-dy}{y-x} = \frac{dy-dz}{z-y} = \frac{dz-dx}{x-z}$$

Multiply  $(-)$  in denominator.

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$$

Case-1:-  $\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$

$$\int \frac{d(x-y)}{(x-y)} = \int \frac{d(y-z)}{y-z}$$

$$\log(x-y) = \log(y-z) + \log c$$

$$\log\left(\frac{x-y}{y-z}\right) = \log c$$

$$\therefore u = \frac{x-y}{y-z}$$

Case-2:- Let  $\frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$

$$\int \frac{d(y-z)}{(y-z)} = \int \frac{d(z-x)}{(z-x)}$$

$$\log(y-z) = \log(z-x) + \log C_2$$

$$\log\left(\frac{y-z}{z-x}\right) = \log C_2$$

$$\frac{y-z}{z-x} = C_2$$

$$v = \frac{y-z}{z-x}$$

$$\phi(u, v) = 0$$

The solution is given by

$$\phi\left[\frac{x-y}{y-z}, \frac{y-z}{z-x}\right] = 0$$



Form the PDE by eliminating the arbitrary function from  $\phi(xy+z^2, x+y+z)=0$ .

Sol: Let  $\phi(xy+z^2, x+y+z)=0$

$$\Rightarrow \phi(u,v)=0$$

$$\therefore u=xy+z^2, v=x+y+z$$

$$\Rightarrow \frac{\partial u}{\partial x} = y + 2z \frac{\partial z}{\partial x} = y + 2pz$$

$$\Rightarrow \frac{\partial u}{\partial y} = x + 2z \frac{\partial z}{\partial y} = x + 2qz$$

$$\Rightarrow \frac{\partial v}{\partial x} = 1 + \frac{\partial z}{\partial x} = 1+p$$

$$\Rightarrow \frac{\partial v}{\partial y} = 1 + \frac{\partial z}{\partial y} = 1+q$$

$$\therefore \text{The required PDE is } \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial y} - \frac{\partial v}{\partial x} \cdot \frac{\partial u}{\partial y} = 0$$

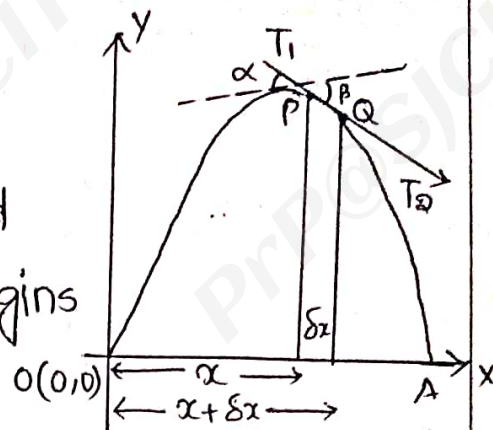
$$\Rightarrow (1+q)(y+2pz) - (1+p)(x+2qz) = 0$$

$$\Rightarrow y + 2pz + qy + 2pqz - x - 2qz - px - 2pqz = 0$$

$$\Rightarrow (y-x) + (qy-px) + 2pz - 2qz = 0.$$

## One-dimensional wave equation:-

Let  $u(x, t)$  be a displacement of a elastic string having the length '1' and it has been stretched two points origins and A.



Let  $T_1, T_2$  be the tensions towards the point P & Q. Let  $\alpha, \beta$  be the angles at the point P and Q at the initial line.

When the string is in a neutral point

$$\therefore T_1 \cos \alpha = T_2 \cos \beta = T$$

and we have the force  $F = -T \sin \alpha + T_2 \sin \beta$

$$\Rightarrow F = T_2 \sin \beta - T_1 \sin \alpha$$

$$\Rightarrow F = \frac{T}{\cos \beta} \sin \beta - \frac{T}{\cos \alpha} \sin \alpha$$

$$\Rightarrow F = T \tan \beta - T \tan \alpha \rightarrow (1)$$

where by the Newton's second law, we have

$F = ma$  where  $m$  is the mass and it is  $m \delta x$

$$\therefore F = m \delta x \frac{\partial^2 u}{\partial t^2}$$

$$\therefore (1) \Rightarrow T \tan \beta - T \tan \alpha = m \delta x \frac{\partial^2 u}{\partial t^2}$$

$$\Rightarrow \tan \beta - \tan \alpha = \frac{m}{T} \delta x \frac{\partial^2 u}{\partial t^2}$$

$$\Rightarrow \left( \frac{\partial u}{\partial x} \right)_{x+\delta x} - \left( \frac{\partial u}{\partial x} \right)_x = \frac{m}{T} \delta x \frac{\partial^2 u}{\partial t^2}$$

$$\Rightarrow \frac{\left( \frac{\partial u}{\partial x} \right)_{x+\delta x} - \left( \frac{\partial u}{\partial x} \right)_x}{\delta x} = \frac{m}{T} \frac{\partial^2 u}{\partial t^2}$$

$$\Rightarrow \lim_{\partial x \rightarrow 0} \frac{\left(\frac{\partial u}{\partial x}\right)_{x+\partial x} - \left(\frac{\partial u}{\partial x}\right)_x}{\partial x} = \frac{m}{T} \frac{\partial^2 u}{\partial t^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{m}{T} \frac{\partial^2 u}{\partial t^2}$$

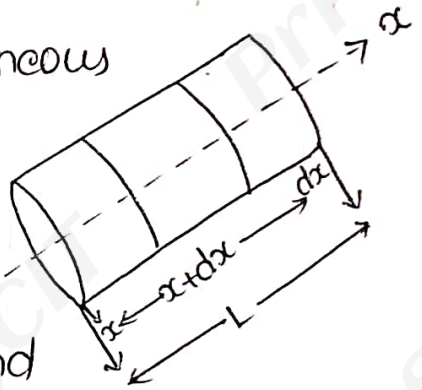
$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = \frac{T}{m} \frac{\partial^2 u}{\partial x^2}$$

$$\Rightarrow \boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}} \quad (\because c^2 = \frac{T}{m})$$

### One dimensional heat equation:-

Consider a heat conducting homogeneous rod of length  $l$  placed along the  $x$ -axis. One end of rod at  $x=0$  (origin) other end of rod at  $x=L$ . Assume that the rod has constant density  $\rho$  and uniform cross section  $A$ . Also assume that the rod is insulated laterally and therefore heat flows only in the  $x$ -direction. Let  $u(x, t)$  with the temperature of the cross section at the point  $x$  & at any time  $t$  the thermal conductivity of  $K$  of the material of the rod on the temperature gradient  $\frac{\partial u}{\partial x}$ .  $q_1$  be the Quantity of heat flowing into the cross section at a distance  $x$  in the unit time is

$q_1 = -KA \left(\frac{\partial u}{\partial x}\right)_x$  and  $q_2$  is the quantity of heat flowing out of the cross section at a distance  $x+dx$





$$q_2 = kA \left( \frac{\partial u}{\partial x} \right)_{x+dx}$$

$$q_1 - q_2 = -kA \left( \frac{\partial u}{\partial x} \right) + kA \left( \frac{\partial u}{\partial x} \right)_{x+dx}$$

$$q_1 - q_2 = kA \left[ \left( \frac{\partial u}{\partial x} \right)_{x+dx} - \left( \frac{\partial u}{\partial x} \right)_x \right] \rightarrow (1)$$

But the rate of increase of heat in the rod is

$$\partial \int A dx = \frac{\partial u}{\partial x}$$

$$q_1 - q_2 = \partial \int A dx \frac{\partial u}{\partial t} \rightarrow (2)$$

$\partial$  is the specific heat

$\rho$  is the density of material

$$\partial \int A dx \frac{\partial u}{\partial t} = kA \left[ \left( \frac{\partial u}{\partial x} \right)_{x+dx} - \left( \frac{\partial u}{\partial x} \right)_x \right]$$

$$\Rightarrow \frac{\partial \rho}{k} \frac{\partial u}{\partial t} = \frac{\left( \frac{\partial u}{\partial x} \right)_{x+dx} - \left( \frac{\partial u}{\partial x} \right)_x}{\partial x}$$

$$\Rightarrow \frac{\delta \rho}{k} \frac{\partial u}{\partial t} = \frac{\lim_{\partial x \rightarrow 0} \left( \frac{\partial u}{\partial x} \right)_{x+\partial x} - \left( \frac{\partial u}{\partial x} \right)_x}{\partial x}$$

$$\Rightarrow \frac{\delta \rho}{k} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$\Rightarrow \frac{\partial u}{\partial t} = \frac{k}{\delta \rho} \frac{\partial^2 u}{\partial x^2}$$

$$\left[ \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \right] \text{ where } c^2 = \frac{k}{\delta \rho}$$

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